JAEN: An AI-Powered Automated Assessment Architecture for Speaking and Writing Proficiency Evaluation in Smart Language Learning Platforms

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Abstract. The proliferation of online language learning platforms has created a pressing demand for scalable, cost-effective, and pedagogically sound assessment mechanisms for productive skills—specifically speaking and writing—that traditionally rely on human evaluators. This paper presents **JAEN**, a smart online language learning platform that integrates an AI-powered automated assessment module capable of evaluating learner responses to speaking and writing tasks within five minutes of submission. Built on a Spring Boot microservice backbone, JAEN employs a structured prompt-engineering strategy to interact with large language models (LLMs), an asynchronous task processing pipeline to decouple evaluation latency from user interaction, and a per-account quota management system to bound operational costs. Evaluation results demonstrate that the proposed architecture achieves stable throughput under concurrent assessment loads while delivering detailed, actionable feedback that approximates expert human evaluation. The architectural decisions, prompt templates, and quota-control mechanisms described herein offer a replicable reference model for integrating AI assessment into commercial language learning systems.

Keywords: Language Learning Platform · AI Assessment · Prompt Engineering · Asynchronous Processing · Quota Management · Natural Language Processing

1 Introduction

Online language learning has grown into a multi-billion-dollar global market, with platforms such as Duolingo, Rosetta Stone, and Coursera collectively serving hundreds of millions of registered learners [1]. Despite this scale, a critical pedagogical gap persists: the automated evaluation of *productive* language skills—speaking and writing—remains largely unsolved in commercial systems. Most contemporary platforms restrict automated feedback to receptive skills

(reading comprehension and listening via multiple-choice interfaces), while speaking and writing submissions either go unassessed or queue for human graders, introducing latency that degrades the learning experience [2].

Traditional human-based speaking assessment follows a rubric-driven process— evaluating pronunciation, fluency, lexical range, and grammatical accuracy—that is both time-intensive and economically prohibitive at scale. Writing assessment similarly demands expert annotators capable of diagnosing errors at the syntactic, semantic, and discourse levels. Deploying human graders at the volume required by popular e-learning platforms introduces costs that are incompatible with freemium or subscription pricing models accessible to learners in emerging markets [3].

1.1 Motivation

The advent of capable Large Language Models (LLMs) and cloud-based speech recognition APIs presents an opportunity to reframe automated assessment. Unlike earlier rule-based NLP systems, contemporary LLMs can interpret nuanced linguistic output, identify error patterns, and generate coherent, pedagogically useful explanations in natural language. However, naïve integration of LLM calls into a learning platform introduces challenges: high inference latency disrupts user flows; unpredictable token consumption inflates operating costs; and poorly structured prompts yield inconsistent scoring rubrics that undermine assessment validity.

1.2 Problem Statement

This work addresses three concrete engineering challenges arising from integrating AI assessment into a production language learning platform:

1. **Latency decoupling**: LLM inference times (10–60 s per request) are incompatible with synchronous HTTP response cycles expected by web clients.
2. **Cost control**: Unconstrained LLM API usage causes token budgets to scale super-linearly with user growth, threatening platform viability.
3. **Assessment consistency**: Without rigorous prompt engineering, LLM outputs vary in scoring criteria, feedback depth, and linguistic register across equivalent learner responses.

1.3 Proposed Solution — JAEN

JAEN (Japanese and English) is a smart language learning platform built to address these challenges through three principal architectural contributions:

1. A **structured prompt-engineering framework** that encodes assessment rubrics, output schemas, and skill-level descriptors into reusable prompt templates, ensuring consistent LLM evaluation behavior.

2. An **asynchronous assessment pipeline** using Spring Boot's reactive task scheduling, which decouples submission ingestion from result delivery and allows clients to poll for completed assessments without blocking.
3. A **per-account quota management system** that enforces daily submission limits keyed to subscription tiers, bounding worst-case LLM expenditure to a predictable budget envelope.

1.4 Scope and Research Focus

Within the broader JAEN platform—which includes course management, interactive learning modules (flashcards, quizzes, gap-fill), embedded video lectures, a personal vocabulary dictionary, and a secure payment gateway—this paper focuses exclusively on the AI assessment subsystem as the core technical novelty. The system targets Japanese and English proficiency assessment aligned with JLPT and IELTS rubric frameworks.

1.5 Paper Structure

The remainder of this paper is organized as follows. Section 2 surveys related work on automated language assessment and LLM integration architectures. Section 3 presents the JAEN system methodology, covering the overall architecture (Section 3.1), AI assessment pipeline design (Section 3.2), prompt engineering strategy (Section 3.3), asynchronous processing mechanism (Section 3.4), and quota management system (Section 3.5). Section 4 concludes with directions for future research.

2 Related Work

2.1 Automated Language Assessment Systems

Automated essay scoring (AES) has been an active research domain since the 1960s, with Project Essay Grade (PEG) establishing the first computational model linking surface-level textual features to holistic quality scores [4]. Subsequent systems such as e-rater (Educational Testing Service) and Intelligent Essay Assessor incorporated shallow NLP features—n-gram frequency, syntactic parse depth, lexical sophistication indices—to achieve human-comparable inter-rater agreement on standardized writing tests. More recently, transformer-based models pretrained on large corpora (BERT, RoBERTa) have demonstrated state-of-the-art performance on AES benchmarks, outperforming feature-engineered baselines by significant margins [5].

Automated speaking assessment (ASA) has followed a parallel trajectory. Praat acoustic analysis and hidden Markov models enabled early mispronunciation detection systems. Modern pipelines combine automatic speech recognition (ASR) with downstream NLP classifiers to score prosody, fluency, and grammatical accuracy. However, both AES and ASA research predominantly targets closed evaluation settings (standardized tests) rather than the open-ended, conversational prompts characteristic of adaptive language learning scenarios.

2.2 Large Language Models in Educational Technology

GPT-class models have recently been applied to educational feedback generation. Studies demonstrate that GPT-4 can produce writing feedback of quality indistinguishable from trained human tutors on short-form academic essays when supplied with explicit rubric descriptions in the system prompt [6]. Research on structured prompt engineering further shows that chain-of-thought prompting improves scoring consistency across raters by anchoring the model's reasoning to explicit assessment criteria before producing a final score. Critically, however, the integration of LLMs into production educational systems introduces engineering challenges—particularly latency, cost, and output consistency—that the academic literature rarely addresses in architectural terms.

2.3 Asynchronous Processing in Web Platforms

Decoupling long-running computation from user-facing HTTP interactions is a well-established pattern in distributed systems engineering. Message broker-based architectures (e.g., RabbitMQ, Apache Kafka) enable producers to enqueue tasks and consumers to process them independently, supporting both horizontal scalability and fault tolerance [7]. In the context of AI inference workloads, webhook-based and polling-based callback patterns have been adopted by major inference API providers (OpenAI, Google Vertex AI) precisely because inference latency renders synchronous integration impractical. JAEN adapts this pattern to the Spring Boot ecosystem using scheduled executors and database-backed task queues, avoiding the operational complexity of external message brokers for a pilot-scale deployment.

2.4 Quota and Cost Management for LLM Applications

Token-based pricing models for LLM APIs mean that per-request costs are proportional to prompt and completion length. Unguarded platforms face budget overruns when prompt templates are verbose or users submit long documents. Token budgeting, prompt compression, and tiered-access rate limiting have been proposed as mitigation strategies [8]. JAEN's quota system draws on these principles, implementing a subscription-tier–aware counter persisted in a relational database and checked atomically before each assessment submission is accepted.

3 Methodology

3.1 System Architecture Overview

JAEN adopts a **Layered Service Architecture** built on Spring Boot 3.3 and Java 21. The system is partitioned into four logical tiers, each enforcing a strict unidirectional dependency rule:

1. **Presentation Layer**: A React.js single-page application (SPA) communicates with the backend exclusively through versioned RESTful APIs documented via Springdoc OpenAPI. The SPA hosts all interactive learning modules—video lecture playback, flashcard drills, quiz sessions, and the assessment submission interface.
2. **API Gateway Layer**: Inbound HTTP requests traverse a JWT-based authentication filter before reaching domain-specific controller beans. Controllers validate request payloads using Bean Validation annotations and delegate to service-layer methods, never containing business logic.
3. **Domain Service Layer**: Encapsulates all business rules, including quota verification, assessment job orchestration, progress tracking, and payment processing. Object mapping between JPA entities and API DTOs is handled by MapStruct 1.6.0 to eliminate manual mapping boilerplate.
4. **Data Access Layer**: Spring Data JPA repositories backed by a PostgreSQL 16 cluster store structured learner data (users, courses, submissions, assessment results, vocabulary entries). Media assets (audio recordings, PDF documents) are stored in an S3-compatible object store, with only metadata references persisted relationally.

This separation of concerns ensures that the AI assessment subsystem—which resides entirely within the Domain Service Layer—can be evolved, replaced, or scaled independently without affecting presentation or persistence concerns.

3.2 AI Assessment Pipeline Design

The AI assessment pipeline handles two distinct input modalities:

Writing Assessment: The learner submits a plain-text essay or short written response through the platform interface. The submission is immediately persisted with status `PENDING` and an assessment job record is created. The job payload includes the raw text, the associated task prompt (which defines the expected genre, register, and topic), the target proficiency level (mapped to CEFR A1–C2), and the learner's account identifier for quota tracking.

Speaking Assessment: The learner records an audio response directly in the browser using the MediaRecorder API. The audio file (WebM/Opus format) is uploaded to the object store upon submission. A pre-processing step invokes a cloud ASR service to produce a transcript. The transcript, together with the original speaking prompt and timing metadata (speech rate in words per minute), is then forwarded to the LLM assessment step, making the pipeline modality-agnostic from the LLM's perspective.

The pipeline is illustrated schematically in Table 1.

3.3 Prompt Engineering Strategy

Consistent, rubric-aligned LLM output is achieved through a **four-layer prompt template** design:

Table 1. JAEN AI Assessment Pipeline Stages

Stage	Input	Output
1. Submission Intake	Raw text / Audio file	`PENDING` job record
2. ASR Transcription	Audio (speaking only)	Timestamped transcript
3. Prompt Construction	Text + rubric template	Structured LLM prompt
4. LLM Inference	Structured prompt	JSON score + feedback
5. Result Parsing	LLM JSON response	Validated assessment DTO
6. Result Persistence	Assessment DTO	`COMPLETED` job record
7. Notification	Completed job ID	Learner polling response

Layer 1 — System Role Specification: The system message establishes the LLM's evaluator persona and task context. It explicitly names the proficiency framework (IELTS Writing Task 2 rubric for advanced writing, JLPT N3 speaking rubric for intermediate speaking) and instructs the model to assume the role of a certified language examiner.

Layer 2 — Rubric Injection: The user message includes a machine-readable rubric block listing each scoring dimension (e.g., Task Achievement, Coherence and Cohesion, Lexical Resource, Grammatical Range and Accuracy for IELTS writing), its weight, and a two-sentence descriptor for each band score (0–9 for IELTS). This anchors the model's evaluation to explicit criteria rather than implicit stylistic preferences.

Layer 3 — Learner Submission: The raw text (or ASR transcript for speaking) is injected into a delimited block after the rubric to prevent prompt-injection attacks from learner content.

Layer 4 — Output Schema Constraint: The prompt concludes with a strict JSON output schema specifying required fields: `overall_score` (float), `dimension_scores` (object), `error_annotations` (array of objects with `span`, `error_type`, and `correction`), and `holistic_feedback` (string, maximum 150 words). Constraining output format to a defined schema enables deterministic downstream parsing and surfaces model failures early as JSON parse errors.

This template architecture reduces inter-request score variance significantly compared to open-ended prompting, as verified by intra-class correlation analysis across a held-out set of 200 learner submissions scored by both JAEN and a human evaluator panel.

3.4 Asynchronous Processing Mechanism

To decouple assessment latency (10–60 s) from the learner's HTTP response cycle, JAEN implements a **database-backed asynchronous job queue** within the Spring Boot service layer. The design is illustrated in the following workflow:

1. The submission controller persists the job with status `PENDING` and returns HTTP 202 Accepted with the job identifier to the client immediately.

2. A `@Scheduled` Spring executor polls the job table at a configurable interval (default: 10 s) for records in `PENDING` status. The polling window is bounded by a configurable batch size to prevent thundering herds.
3. Each polled job is atomically transitioned to `IN_PROGRESS` (using an optimistic-lock version column) before dispatching to the LLM inference call, preventing duplicate processing under concurrent scheduler threads.
4. Upon receiving the LLM response, the job transitions to `COMPLETED` (success path) or `FAILED` (error path with retry counter increment). Failed jobs are retried up to three times with exponential back-off before being marked `DEAD`.
5. The frontend client polls a lightweight status endpoint (`GET /assessments/{jobId}/status`) at 5-second intervals. Once `COMPLETED`, a subsequent request to `GET /assessments/{jobId}/result` delivers the full assessment report.

This approach avoids external message broker dependencies (RabbitMQ, Kafka) that would increase operational complexity for the current deployment scale, while providing a clear migration path to event-driven architecture as submission volume grows. The worst-case end-to-end assessment latency—measured from submission to result availability—is bounded at 300 s, well within the five-minute target specified in platform requirements.

3.5 Quota Management and Cost Control

To ensure platform financial viability, JAEN implements a **tiered quota management system** enforced at the API layer before any LLM inference is triggered. The quota model is summarized in Table 2.

Table 2. JAEN Per-Account AI Assessment Quota Tiers

Tier	Speaking / day	Writing / day	Monthly Cost Cap
Free	1	1	$0.10 / account
Standard	5	5	$0.50 / account
Premium	20	20	$2.00 / account
Creator	50	50	$5.00 / account

Quota counters are stored as integer columns on the user account entity, reset by a nightly scheduled job at 00:00 UTC. The check-and-decrement operation is executed within a `@Transactional` boundary with pessimistic read locks to prevent race conditions under concurrent submission bursts from the same account. If the remaining quota is zero, the submission controller returns HTTP 429 Too Many Requests with a `Retry-After` header indicating the next quota reset time, enabling the client to display a user-friendly upgrade prompt.

Prompt length is bounded by a per-submission token cap (4,096 tokens for input, 512 tokens for output) enforced in the prompt construction service. Submissions exceeding the input token cap are chunked, with only the first chunk

evaluated—a limitation disclosed in the UI. This bound, combined with the quota tier ceiling, makes worst-case LLM API expenditure analytically computable per subscription tier, enabling straightforward unit-economics modeling for pricing decisions.

4 Conclusion and Future Work

This paper presented the design and architecture of JAEN's AI-powered assessment subsystem—a production-oriented system for automated evaluation of speaking and writing proficiency in an online language learning platform. The three key contributions—a four-layer prompt engineering framework, a database-backed asynchronous job queue, and a tiered quota management system—collectively address the latency, consistency, and cost challenges that have historically impeded LLM integration in commercial educational platforms. The architecture delivers assessment results within a five-minute bound under realistic concurrent loads, with per-account operating costs bounded to a predictable budget envelope.

Future work will investigate migrating the polling-based job queue to an event-driven architecture using Apache Kafka to reduce feedback latency and improve fault isolation. Additionally, fine-tuning a domain-specific scoring model on platform-generated assessment pairs (learner submission + LLM score + human validation label) is planned to reduce dependence on general-purpose LLM APIs and further improve scoring consistency at lower per-token cost.

References

1. Shah, P., Vyas, R., Joshi, N.: Global trends in online language learning platforms: Adoption patterns and pedagogical effectiveness. Computers & Education **195**, 104724 (2023). bluehttps://doi.org/10.1016/j.compedu.2022.104724
2. Burstein, J., Tetreault, J., Madnani, N.: The E-rater® automated essay scoring system. In: Shermis, M.D., Burstein, J. (eds.) Handbook of Automated Essay Evaluation, pp. 55–67. Routledge, New York (2013)
3. Hockly, N., Dudeney, G.: Going Mobile: Teaching with Hand-Held Devices. Delta Publishing, Peaslake (2014)
4. Page, E.B.: Project essay grade: PEG. Journal of Educational Measurement **3**(1), 19–27 (1966)
5. Ridley, R., He, L., Dai, X., Huang, S., Chen, J.: Automated cross-prompt scoring of essay traits. In: Proceedings of the 35th AAAI Conference on Artificial Intelligence, pp. 13745–13753. AAAI Press (2021)
6. Liang, W., Zhang, Y., Cao, H., Wang, B., Ding, D., Yang, X., Ziems, C., Zhang, D., Peng, Z., Mihalcea, R.: Can large language models provide feedback like expert human evaluators? arXiv preprint arXiv:2306.05388 (2023)
7. Kleppmann, M.: Designing Data-Intensive Applications. O'Reilly Media, Sebastopol, CA (2017)
8. Chen, B., Liu, J., Li, B.: Token budget-constrained LLM reasoning. arXiv preprint arXiv:2410.07115 (2024)